

Contents

Preface	<i>xvii</i>
Series Preface	<i>xxi</i>
Acknowledgments	<i>xxiii</i>
List of Acronyms	<i>xxv</i>
About the Companion Website	<i>xxix</i>

Part I Introduction 1

1	History and Overview	3
1.1	Overview	3
1.2	History	4
1.2.1	Early History	4
1.2.2	The Vietnam War	4
1.2.3	Resurgence	5
1.2.4	Joint Operations	6
1.2.5	Desert Storm	6
1.2.6	Bosnia	6
1.2.7	Afghanistan and Iraq	7
1.2.8	Long-Range Long-Endurance Operations	7
1.3	Overview of UAV Systems	8
1.3.1	Air Vehicle	9
1.3.2	Mission Planning and Control Station	9
1.3.3	Launch and Recovery Equipment	10
1.3.4	Payloads	10
1.3.5	Data Links	11
1.3.6	Ground Support Equipment	11
1.4	The Aquila	12
1.4.1	Aquila Mission and Requirements	12
1.4.2	Air Vehicle	13
1.4.3	Ground Control Station	13
1.4.4	Launch and Recovery	14
1.4.5	Payload	14
1.4.6	Other Equipment	14
1.4.7	Summary	14

1.5	Global Hawk	16
1.5.1	Mission Requirements and Development	16
1.5.2	Air Vehicle	16
1.5.3	Payloads	17
1.5.4	Communications System	17
1.5.5	Development Setbacks	18
1.6	Predator Family	19
1.6.1	Predator Development	19
1.6.2	Reaper	19
1.6.3	Features	20
1.7	Top UAV Manufacturers	21
1.8	Ethical Concerns of UAVs	21
	Questions	22

2 Classes and Missions of UAVs 27

2.1	Overview	27
2.2	Classes of UAV Systems	27
2.2.1	Classification Criteria	27
2.2.2	Classification by Range and Endurance	28
2.2.3	Classification by Missions	29
2.2.4	The Tier System	32
2.3	Examples of UAVs by Size Group	33
2.3.1	Micro-UAVs	34
2.3.2	Mini-UAVs	34
2.3.3	Very Small UAVs	35
2.3.4	Small UAVs	36
2.3.5	Medium UAVs	37
2.3.6	Large UAVs	40
2.4	Expendable UAVs	42
	Questions	43

Part II The Air Vehicle 47

3 Aerodynamics 49

3.1	Overview	49
3.2	Aerodynamic Forces	49
3.3	Mach Number	51
3.4	Airfoil	51
3.5	Pressure Distribution	54
3.6	Drag Polar	57
3.7	The Real Wing and Airplane	58
3.8	Induced Drag	58
3.9	Boundary Layer	60
3.10	Friction Drag	63
3.11	Total Air-Vehicle Drag	63

3.12	Flapping Wings	64
3.13	Aerodynamic Efficiency	66
	Questions	67
4	Performance	71
4.1	Overview	71
4.2	Cruising Flight	72
4.3	Range	73
	4.3.1 Range for a Non-Electric-Engine Propeller-Driven Aircraft	74
	4.3.2 Range for a Jet-Propelled Aircraft	76
4.4	Endurance	78
	4.4.1 Endurance for a Non-Electric-Engine Propeller-Driven Aircraft	78
	4.4.2 Endurance for a Jet-Propelled Aircraft	79
4.5	Climbing Flight	80
4.6	Gliding Flight	82
4.7	Launch	83
4.8	Recovery	84
	Questions	85
5	Flight Stability and Control	89
5.1	Overview	89
5.2	Trim	90
	5.2.1 Longitudinal Trim	91
	5.2.2 Directional Trim	93
	5.2.3 Lateral Trim	93
	5.2.4 Summary	94
5.3	Stability	94
	5.3.1 Longitudinal Static Stability	95
	5.3.2 Directional Static Stability	97
	5.3.3 Lateral Static Stability	99
	5.3.4 Dynamic Stability	100
5.4	Control	101
	5.4.1 Aerodynamic Control	101
	5.4.2 Pitch Control	102
	5.4.3 Directional Control	104
	5.4.4 Lateral Control	105
	Questions	106
6	Propulsion	109
6.1	Overview	109
6.2	Propulsion Systems Classification	109
6.3	Thrust Generation	110
6.4	Powered Lift	111
6.5	Sources of Power	114
	6.5.1 Four-Cycle Engine	115
	6.5.2 Two-Cycle Engine	116

6.5.3	Rotary Engine	118
6.5.4	Gas Turbine Engines	119
6.5.5	Electric Motors	120
6.6	Sources of Electric Energy	122
6.6.1	Batteries	122
6.6.2	Solar Cells	124
6.6.3	Fuel Cells	126
6.7	Power and Thrust	127
6.7.1	Relation Between Power and Thrust	128
6.7.2	Propeller	128
6.7.3	Variations of Power and Thrust with Altitude	130
	Questions	131

7	Air Vehicle Structures	135
7.1	Overview	135
7.2	Structural Members	135
7.2.1	Skin	136
7.2.2	Fuselage Structural Members	136
7.2.3	Wing and Tail Structural Members	137
7.2.4	Other Structural Members	138
7.3	Basic Flight Loads	139
7.4	Dynamic Loads	143
7.5	Structural Materials	145
7.5.1	Overview	145
7.5.2	Aluminum	145
7.6	Composite Materials	146
7.6.1	Sandwich Construction	146
7.6.2	Skin or Reinforcing Materials	147
7.6.3	Resin Materials	147
7.6.4	Core Materials	148
7.7	Construction Techniques	148
7.8	Basic Structural Calculations	149
7.8.1	Normal and Shear Stress	150
7.8.2	Deflection	152
7.8.3	Buckling Load	153
7.8.4	Factor of Safety	154
7.8.5	Structural Fatigue	155
	Questions	155

Part III Mission Planning and Control 159

8	Mission Planning and Control Station	161
8.1	Introduction	161
8.2	MPCS Subsystems	161
8.3	MPCS Physical Configuration	165

8.4	MPCS Interfaces	169
8.5	MPCS Architecture	170
8.5.1	Fundamentals	170
8.5.2	Local Area Networks	172
8.5.3	Levels of Communication	172
8.5.4	Bridges and Gateways	173
8.6	Elements of a LAN	174
8.6.1	Layout and Logical Structure (Topology)	174
8.6.2	Communications Medium	175
8.6.3	Network Transmission and Access	175
8.7	OSI Standard	175
8.7.1	Physical Layer	175
8.7.2	Data-Link Layer	175
8.7.3	Network Layer	176
8.7.4	Transport Layer	176
8.7.5	Session Layer	176
8.7.6	Presentation Layer	176
8.7.7	Application Layer	176
8.8	Mission Planning	176
8.9	Pilot-In-Command	179
	Questions	180
9	Control of Air Vehicle and Payload	183
9.1	Overview	183
9.2	Levels of Control	184
9.3	Remote Piloting the Air Vehicle	185
9.3.1	Remote Manual Piloting	186
9.3.2	Autopilot-Assisted Control	188
9.3.3	Complete Automation	188
9.3.4	Summary	189
9.4	Autopilot	190
9.4.1	Fundamental	190
9.4.2	Autopilot Categories	191
9.4.3	Inner and Outer Loops	191
9.4.4	Overall Modes of Operation	192
9.4.5	Control Process	193
9.4.6	Control Axes	193
9.4.7	Controller	194
9.4.8	Actuator	195
9.4.9	Open-Source Commercial Autopilots	195
9.5	Sensors Supporting the Autopilot	196
9.5.1	Altimeter	196
9.5.2	Airspeed Sensor	196
9.5.3	Attitude Sensors	197
9.5.4	GPS	198
9.5.5	Accelerometers	199

- 9.6 Navigation and Target Location 199
- 9.7 Controlling Payloads 201
 - 9.7.1 Signal Relay Payloads 202
 - 9.7.2 Atmospheric, Radiological, and Environmental Monitoring 202
 - 9.7.3 Imaging and Pseudo-Imaging Payloads 203
- 9.8 Controlling the Mission 204
- 9.9 Autonomy 206
 - Questions 208

Part IV Payloads 213

- 10 Reconnaissance/Surveillance Payloads 215**
 - 10.1 Overview 215
 - 10.2 Imaging Sensors 216
 - 10.3 Target Detection, Recognition, and Identification 217
 - 10.3.1 Sensor Resolution 218
 - 10.3.2 Target Contrast 221
 - 10.3.3 Transmission Through the Atmosphere 222
 - 10.3.4 Target Signature 225
 - 10.3.5 Display Characteristics 225
 - 10.3.6 Range Prediction Procedure 226
 - 10.3.7 A Few Considerations 228
 - 10.3.8 Pitfalls 230
 - 10.4 The Search Process 231
 - 10.4.1 Types of Search 231
 - 10.4.2 Field of View 232
 - 10.4.3 Search Pattern 234
 - 10.4.4 Search Time 235
 - 10.5 Other Considerations 237
 - 10.5.1 Location and Installation 237
 - 10.5.2 Stabilization of the Line of Sight 238
 - 10.5.3 Gyroscope and Gimbal 238
 - 10.5.4 Gimbal-Gyro Configuration 240
 - 10.5.5 Thermal Design 241
 - 10.5.6 Environmental Conditions Affecting Stabilization 241
 - 10.5.7 Boresight 242
 - 10.5.8 Stabilization Design 242
 - Questions 243
- 11 Weapon Payloads 247**
 - 11.1 Overview 247
 - 11.2 History of Lethal Unmanned Aircraft 249
 - 11.3 Mission Requirements for Armed Utility UAVs 251
 - 11.4 Design Issues Related to Carriage and Delivery of Weapons 252
 - 11.4.1 Payload Capacity 253
 - 11.4.2 Structural Issues 253

11.4.3	Electrical Interfaces	255
11.4.4	Electromagnetic Interference	256
11.4.5	Launch Constraints for Legacy Weapons	257
11.4.6	Safe Separation	257
11.4.7	Data Links	258
11.4.8	Payload Location	258
11.5	Signature Reduction	258
11.5.1	Acoustical Signatures	259
11.5.2	Visual Signatures	263
11.5.3	Infrared Signatures	264
11.5.4	Radar Signatures	265
11.5.5	Emitted Signals	269
11.5.6	Active Susceptibility Reduction Measures	269
11.6	Autonomy for Weapon Payloads	270
11.6.1	Fundamental Concept	270
11.6.2	Rules of Engagement	272
	Questions	273
12	Other Payloads	277
12.1	Overview	277
12.2	Radar	277
12.2.1	General Radar Considerations	277
12.2.2	Synthetic Aperture Radar	280
12.3	Electronic Warfare	280
12.4	Chemical Detection	281
12.5	Nuclear Radiation Sensors	282
12.6	Meteorological and Environmental Sensors	282
12.7	Pseudo-Satellites	283
12.8	Robotic Arm	286
12.9	Package and Cargo	287
12.10	Urban Air Mobility	288
	Questions	288

Part V Data Links 291

13	Data Link Functions and Attributes	293
13.1	Overview	293
13.2	Background	293
13.3	Data-Link Functions	295
13.4	Desirable Data-Link Attributes	296
13.4.1	Worldwide Availability	297
13.4.2	Resistance to Unintentional Interference	298
13.4.3	Low Probability of Intercept (LPI)	298
13.4.4	Security	298
13.4.5	Resistance to Deception	299
13.4.6	Anti-ARM	299

13.4.7	Anti-Jam	300
13.4.8	Digital Data Links	301
13.4.9	Signal Strength	301
13.5	System Interface Issues	302
13.5.1	Mechanical and Electrical	302
13.5.2	Data-Rate Restrictions	302
13.5.3	Control-Loop Delays	303
13.5.4	Interoperability, Interchangeability, and Commonality	305
13.6	Antennas	307
13.6.1	Omnidirectional Antenna	307
13.6.2	Parabolic Reflectors	307
13.6.3	Array/Directional Antennas	308
13.6.4	Lens Antennas	308
13.7	Data Link Frequency	309
	Questions	311
14	Data-Link Margin	315
14.1	Overview	315
14.2	Sources of Data-Link Margin	316
14.2.1	Transmitter Power	316
14.2.2	Antenna Gain	316
14.2.3	Processing Gain	322
14.3	Anti-Jam Margin	327
14.3.1	Definition of Anti-Jam Margin	327
14.3.2	Jammer Geometry	328
14.3.3	System Implications of AJ Capability	331
14.3.4	Anti-Jam Uplinks	334
14.4	Propagation	334
14.4.1	Obstruction of the Propagation Path	334
14.4.2	Atmospheric Absorption	336
14.4.3	Precipitation Losses	336
14.5	Data-Link Signal-to-Noise Budget	336
	Questions	339
15	Data-Rate Reduction	343
15.1	Overview	343
15.2	Compression Versus Truncation	343
15.3	Video Data	344
15.3.1	Gray Scale	344
15.3.2	Encoding of Gray Scale	345
15.3.3	Effects of Bandwidth Compression on Operator Performance	346
15.3.4	Frame Rate	348
15.3.5	Control Loop Mode	348
15.3.6	Forms of Truncation	350
15.3.7	Summary	351
15.4	Non-Video Data	352
15.5	Location of the Data-Rate Reduction Function	353
	Questions	354

16 Data-Link Tradeoffs 357

- 16.1 Overview 357
- 16.2 Basic Tradeoffs 357
- 16.3 Pitfalls of “Putting Off” Data-Link Issues 359
- 16.4 Future Technology 360
- Questions 360

Part VI Launch and Recovery 363**17 Launch Systems 365**

- 17.1 Overview 365
- 17.2 Conventional Takeoff 365
- 17.3 Basic Considerations 366
- 17.4 Launch Methods for Fixed-Wing Air Vehicles 370
 - 17.4.1 Overview 370
 - 17.4.2 Rail Launchers 372
 - 17.4.3 Pneumatic Launchers 373
 - 17.4.4 Hydraulic-Pneumatic Launchers 374
 - 17.4.5 Zero Length RATO Launch of UAVs 375
 - 17.4.6 Tube Launch 375
- 17.5 Rocket-Assisted Takeoff 376
 - 17.5.1 RATO Configuration 376
 - 17.5.2 Ignition Systems 376
 - 17.5.3 Expended RATO Separation 376
 - 17.5.4 Other Launch Equipment 377
 - 17.5.5 Energy (Impulse) Required 377
 - 17.5.6 Propellant Weight Required 378
 - 17.5.7 Thrust, Burning Time, and Acceleration 379
- 17.6 Vertical Takeoff 379
- Questions 379

18 Recovery Systems 383

- 18.1 Overview 383
- 18.2 Conventional Landing 383
- 18.3 Vertical Net Systems 384
- 18.4 Parachute Recovery 385
- 18.5 VTOL UAVs 386
- 18.6 Mid-Air Retrieval 388
- 18.7 Shipboard Recovery 389
- 18.8 Break-Apart Landing 392
- 18.9 Skid and Belly Landing 393
- 18.10 Suspended Cables 394
- Questions 395

19 Launch and Recovery Tradeoffs 397

- 19.1 UAV Launch Method Tradeoffs 397
- 19.2 Recovery Method Tradeoffs 400

19.3	Overall Conclusions	402
	Questions	402
20	Rotary-Wing UAVs and Quadcopters	405
20.1	Overview	405
20.2	Rotary-Wing Configurations	406
20.2.1	Single Rotor	406
20.2.2	Twin Co-axial Rotors	407
20.2.3	Twin Tandem Rotors	408
20.2.4	Multicopters	408
20.3	Hybrid UAVs	409
20.3.1	Tilt Rotor	409
20.3.2	Tilt Wing	409
20.3.3	Thrust Vectoring	410
20.3.4	Fixed-Wing Quadcopter Combination	410
20.4	Quadcopters	411
20.4.1	Overview	411
20.4.2	Aerodynamics	413
20.4.3	Control	414
	Questions	416
	References	419
	Index	421